



ICT International

Product Support Documentation

Version: 2.6.2

Release Date: March 2026

Information Resources for new Datalogger/Node Platform (DNP)

DNP information

Datalogger first with integrated IoT communication options. Robust data collection and storage is prioritised, along with user-friendly installation and configuration.

The node has multiple inputs that allow for:

- Easy to use SDI-12 and RS485 connections with:
 - 8x SDI-12 inputs
 - 4x RS485 inputs (and therefore outputs as well that are suitable for control systems)
 - 3x Voltage inputs, capable of 4-20mA and differential measurements
- 1x digital counter (compatible with double reed switch rain gauges)
- Expansion options that allow for single cable expansion using additional measurement modules
 - CANBUS
 - Single Pair

The enclosure has been designed to be as easy to use as possible and for simple installation and includes simple pole mount bracket that allows for easy installation. By using pre-wired connectors, all sensors become plug and play. The datalogger has:

- 4x SDI-12 Connectors, each capable of connecting two sensors
- 2x RS485, each capable of connecting two sensors
- 3x Voltage Connectors
- 1x Digital
- 1x 3 Pin power (solar or benchtop)
- 1x CANBUS

With power provided to the node by either a rechargeable battery and solar panel with either 10, 20 or 40W solar panel options, or benchtop power options for laboratory installations (retaining the internal battery, creating a mini-UPS equipped logger), the node is easy to install without any specialised skills.

With an option to connect an extra solar panel, as well as three sizes of internal rechargeable batteries, the power system is designed to be robust, reliable and adaptable.

Whilst designed as a datalogging platform first, the communication options allow for installation in many remote locations.

New Communication Options

The new node will be available with the following communication options:

- Point to Point WiFi and Bluetooth for configuration/short range data access
- LoRaWAN: All common bands
- Cellular
- Satellite:
 - Iridium SMT internal
 - Iridium CERTUS 100 (via ethernet and external module)

Combined with the ICT International DB2 platform, designed for the visualisation and storage of timeseries data related to soil, plant, and environmental measurements, the Node is a powerful measurement tool for many applications.

Key Features and Benefits

Feature	Benefit
SDI-12	- Designed with 4x ports, each supporting 2 SDI-12 input busses \n- Port-level addressing removes the need to address sensors in many applications \n- Individually controlled power to each SDI-12 input \n- Independent functionality \n- Improved independence if sensor fails \n- Faster configuration \n- No need to address sensors
RS485	- Designed with 2x ports each supporting 2x RS485 channels \n- Individually controlled power to each port \n- Ability to control (transmit commands) other systems \n- Independent functionality and improved independence if sensor fails \n- Faster configuration, no need to address sensors \n- Compatibility with other RTUs and control systems
Digital	- Compatible with double reed switch rain gauges
Voltage	- Differential measurements
4-20mA	- Available with 24-volt power

Technical details New Node/Sensor Package Build List

All tiers for the multiple sensor option would comprise the following:

- Logger: the new Definium Logger, including mounting system
 - Main Board with SD Card Storage
 - Input Connectors
 - 2x RS485
 - 4x SDI-12
 - 3x Voltage

- 1x Digital
- 1x 3 pin Power
- 1x CANBUS
- Takachi Enclosure with mounting bracket
 - Enclosure WG15-21-9W
 - Mounting Bracket: WPMB-M4-2 G
 - Gore-Tex Vent
- Battery, Solar Panel, and mounting system to suit:
 - Low power (LP) using 10W solar and 6B battery
 - 10-Watt Solar Panel
 - Cable
 - Connector
 - 6B Battery
 - Mount (RF Industries Mount)
 - High Power (HP) using 20W solar and 13B battery
 - 20-Watt Solar Panel
 - Cable
 - Connector
 - 13B Battery
 - Mount (RF Industries Mount)
 - Ultra-High Power (UHP) using 40W solar and 26B battery
 - 40-Watt Solar Panel
 - Cable
 - Connector
 - 26B Battery
 - Mount (RF Industries Mount)
 - Extra Power (EP-L/-H/-U)
 - 2nd matched solar panel
 - Cable
 - Connector
 - Diodes
 - Mount to suit
- Mounting system for Sensors (Crossarm) when included
- All connectors and wiring for the sensors

For those that require data connectivity, then there would be the option to include:

- LoRaWAN (-L)
 - Comms board
 - Antenna
 - Integration to TTN
 - Decoder
- Cat-M1 (-C)
 - Comms Module

- Antenna
- 1NCE SIM Card
- Integration to MQTT broker
- Decoder
- 3G/4G/5G (-G)
 - Comms Module
 - Antenna
 - Integration to MQTT broker
 - Decoder
 - *Locally sourced SIM Card*
- Satellite (-S)
 - Comms Module
 - Antenna
 - Subscription for GroundControl IMT
 - Integration to MQTT broker
 - Decoder

Datalogger/Node Platform outside of the packages

When configuring a Standalone ICT International DNP, the following options need to be considered:

- Communications
 - DNP-D: Datalogger only, no IoT Comms
 - DNP-L: LoRaWAN (BOM)
 - DNP-C: Cellular (BOM)
 - DNP-S: Internal Satellite Module (BOM)
- Power Options for DNP
 - LP: Low Power with 6,000mAh battery, 10W solar and mount suit (BOM)
 - HP: High Power with 13,000mAh battery, 20W solar and mount suit (BOM)
 - UHP: Ultra-high power with 26,000mAh battery, 40W solar and mount suit (BOM)
 - Extra Power (EP-LP/-HP/-UHP) comprising a 2nd matched solar panel with: (BOMx3)
 - Cable
 - Connector
 - Diodes
 - Mount to suit

Dashboard and Databasing for the new combinations

By moving to a standardised set of measurement bundles the construction of dashboards for customers will be much simplified and more scalable as an

implementation solution. Each measurement bundle will require a maximum of three dashboards; however, all the parameters are very similar in the data type that will come in. Dashboards can be combined with additional modules to enhance the data collected. These modules are (generally) specific to single sensor (type) measurements on a single node or CANBUS expansion.

Measurement Bundle	Dashboard or Module
Plant water use	Dashboard
SPAC Water Potential	Dashboard
Irrigation Scheduling	Dashboard
Weighing Lysimeter	Dashboard
Weather Station (ET) (Evapotranspiration)	Dashboard
Weather Station (Fire)	Dashboard
Weather Station (Air Quality)	Dashboard
Water Quality	Dashboard
Soil Measurements	Dashboard or Module
Plant Water Use (PWU-E1 or PWU-E2)	Module
Plant Water Potential	Module
Soil Water Potential	Module
Soil Measurements (Multi)	Module
SFMx Sap Flow/SDI-12	Module
Plant Health/CWSI	Module

Measurement Problem Sensor Combinations (priced without datalogger)

Replacing the SNIps, these large system combinations are intended to cover most eventualities that we encounter, whilst providing clear answers to the measurement problems that customers present. When quoting a System from the new list, the only thing to change is the -D, -L, -C, -S, or -3P. A -3P will require power system to be designed as this deletes the power option from the rest of the build.



ICT International

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MP406

Test file



ICT International

Product Support Documentation

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Sap Flow Sensor (SDI-12)

About this document

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Version information

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Related manuals

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Documentation conventions

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Product overview

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Components (what's in the box)

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Optional features

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USB cables

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Installation Kit

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Config Tool

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Backward compatibility

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ESD protection

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Functional description

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Measurement Principle

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Inputs

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Outputs

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Installation

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Site Selection

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Unpacking

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Installing

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Placing

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Mounting

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Aligning

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Configuring offset

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Installing Configuration Tool

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Installing USB cable driver

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Powering the sensor/instrument/device

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Power supplies

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Power management

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Wiring

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Solar

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Connections

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Communication Options

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Comm protocols

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Connection cables

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Communications Settings

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Commands

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exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

SDI-12

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RS-485

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Analogue

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Settings

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Checking the settings

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Setting fields

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Changing the settings

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Maintenance

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Cleaning

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Calibration

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Troubleshooting

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Error messaging

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Missing readings and error indication

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Technical specifications

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Specifications

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Options and accessories

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Networking

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Connecting several sensors

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SDI-12

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RS-485

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Analogue

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exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Decoder -----

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External sensors

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Complimenting sensors

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ICT International

Product Support Documentation

Version: 2.6.3

Release Date: March 2026

Test

Added and edited this file on GitHub web to test the deployment.

Bug fixed for CI pipeline for the failure to create pdf for this file.

Bug fixed for CI pipeline for pdf rendering and toc.yml.

Fixed to stabilize PDF render in GitHub Actions

corrected PDF browser timeout detection to avoid false timeout failures

11:01



ICT International

Product Support Documentation

Version: 2.6.2

Release Date: March 2026

About this document

Version information

Document code/Version	Date	Description

Related manuals

Product Code	Product Name	Brief description
MP406		
DBV60		
DPV40		
SMT100a		
Therm-EP		
PRP-02		

Documentation conventions

 **WARNING**
Dangerous certain consequences of an action.

 **CAUTION**
Negative potential consequences of an action.

 **IMPORTANT**
Essential information required for user success.

NOTE

Information the user should notice even if skimming.

TIP

Optional information to help a user be more successful.

Product overview

The AD-Node is a LoRaWAN node, supplied with either EU868 (Europe), AU915 (Australia), US915 (United States), or AS923 (Asia) frequency plans depending on customer requirements.

The AD-Node is an Internet of Things (IoT) node that is used for the measurement of up to:

- 2x single ended voltage inputs
- 1x 4-20mA input
- 2x Thermistor Input
- 4x Digital Input

With LoRaWAN connectivity, the AD-Node is ideal for a low-power IoT node is needed for analogue sensors.

Powered by internal batteries (3x AA), the AD-Node does not require external solar power.

Functional description

The AD-Node measures voltage, current, and counter inputs using a 24-bit ADC.

Inputs: Sensor

- Voltage (0-3.0 volts)
- Current (4-20mA)
- Digital counter
- Thermistor value

Outputs: Data

The data is output as a single hexadecimal payload, containing all the input variables.

The hexadecimal data requires a decoder at the LoRaWAN Server.

Specific decoders can be found on the ICT International GitHub library.

Components (what's in the box)

Included with the AD-Node are:

- 3x AA batteries
- antenna to suit

Optional features

Options include:

- Node mounting bracket, suitable for poles up to 60 mm in diameter
- Additional sensors

USB cables

To access the node for configuration, a USB 2.0 Type B cable is required (not included).

Config Tool

The AD-Node is shipped pre-configured, ready to go.

Certain LoRaWAN parameters can be changed using Over-The-Air configuration.

Unpacking

The node will be shipped with the sensors connected directly to the node. If the node mounting bracket has been optioned, the node will be attached to the bracket.

⊗ CAUTION

- ESD protection
- Only power the node on once the antenna is attached
- Once sensors are connected, confirm polarity before turning the power on.

The node board is protected with a conformal coating, which reduces the risk of Electrostatic Discharge (ESD).

Preparing the AD-Node for use

⊗ CAUTION

- Do not change any of the LoRaWAN key settings unless you are changing the gateway connection of the device.
- Changes made to LoRaWAN keys will require reprogramming at the LoRaWAN Server

⊗ IMPORTANT

Site Selection:

When choosing the site for the device installation, consideration must be given to:

- Gateway location
- Node positioning
- Antenna orientation
- Obstructions between the node and gateway

Installation location

The installation location of the AD-Node is determined by:

- Signal strength
- Sensor positions
- Cable lengths

Positioning of the AD-Node

Position the AD-Node towards the LoRaWAN gateway to optimise the signal.

Ensure that there are no obstructions within the following distances:

LoRaWAN Frequency	Whole wavelength in centimetres	Minimum distance to obstruction
EU868 (Europe)	34.5	17 cm
AU915 (Australia)	32.7	16 cm
US915 (United States)	32.7	16 cm
AS923 (Asia)	32.4	16 cm

Mounting the AD-Node

Included with the AD-Node are 4 brackets that can be screwed to the rear of the node enclosure using the moulded holes in the corners. These can be used to mount the node enclosure wall or post.



Powering the device

The AD-Node is a self-powered device with 3x AA batteries. It is recommended to use 1.5-volt lithium batteries when replacements are needed. The supplied AA batteries are L91 Energizer single use batteries, mAh, 1.5 volt.

To power the device on, move the switch to the On position.

The unit will start to transmit at the logging interval from the moment it is powered on. This is confirmed by the LED light sequence:

Red, blue sparkles

Connecting sensors

The AD-Node is supplied with sensors connected. If the sensors need to be disconnected (e.g. for running cables through conduit) then the following needs to be followed.

Each connector in the enclosure is almost tool-less for installation, with a push to fit connector.

To release the cable, a jeweller's flat head screwdriver is required.

Configuration Changes

Communication Settings

Any changes that are needed to be made, such as measurement frequency, can be made using the node terminal function.

Node Terminal is downloadable from:

Checking the settings

Settings can be checked using Node Terminal

⊗ CAUTION

- Do not change any of the LoRaWAN key settings unless you are changing the gateway connection of the device.
- Changes made to LoRaWAN keys will require reprogramming at the LoRaWAN Server

Setting LoRaWAN fields

The settings fields that can be changed are:

- LoRaWAN Keys
- Sensor/Measurement Settings

Changing the LoRa settings

To retrieve or reprogram LoRaWAN keys (OTAA), enter the following:

To Retrieve:	To Reprogram:
lora eui dev	lora eui dev 0011223344556677
lora eui app	lora eui app 0011223344556677
lora key app	lora key app 00112233445566778899887766554433

To Retrieve:	To Reprogram:
lora eui dev	lora eui dev 0011223344556677
lora eui app	lora eui app 0011223344556677
lora key app	lora key app 00112233445566778899887766554433

Sensor Settings

Changing the measurement frequency

The upload schedule can be changed using the command report period. By default the system will be shipped at a 10-minute logging interval (600).

e.g. changing to a 15-minute interval would use the following code:

report period 900

Analogue Configuration

Analogue measurements can be post-processed in the cloud

Maintenance

Maintenance is at the sensor level, rather than the device level.

The batteries are replaceable. To replace follow the steps below:

- Open the enclosure
- Turn the power off by moving the switch to the 1 position
- Release the battery retaining strap
- Remove the batteries
- Check the polarity of the new batteries
- Insert the batteries
- Replace the retaining strap
- Turn the device switch to the On position
- Clean the rubber seal and opposite surface
- Close the enclosure, and ensure that both latches are fully closed

Calibration

There is no calibration of the device available.

Calibration is undertaken on a sensor-by-sensor basis.

Troubleshooting

Plain text

Error messaging

Plain text

Missing readings and error indication

Plain text

Technical specifications

All technical specifications are subject to change as the products undergo continuous improvement

Specifications

	Detail
Analogue	Analogue inputs with selectable voltage 0 - 12v Input Range 4x Single Ended or 2x Differential
Digital	Digital inputs with pulse counting Up to 1kHz input frequency
Operating Environment	-20°C to 60°C
Enclosure Dimensions:	IP65 Clam-Shell 140 ×120 ×80 mm, 500 g
Certifications:	AS/NZS 60950.1:2011 AS/NZS 4268:2012 RoHS compliant (lead-free)

Options and accessories

Plain text

Compatible sensors

Product Code	Product Name	Maximum possible number*
MP406		
DBV60		
DPV40		
SMT100a		
Therm-EP		
PRP-02		

* this is the maximum number of that sensor that can on an individual node.
Combinations will be different.



ICT International

Product Support Documentation

Version: 2.6.4

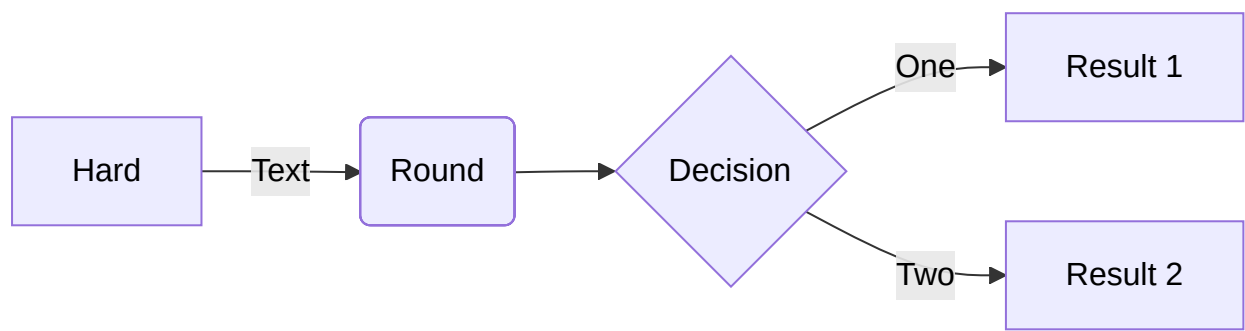
Release Date: March 2026

MFR Node

This is a test file created to check if the system (render, build and injection) works for the added content.

The Cauchy-Schwarz Inequality

$$\left(\sum_{k=1}^n a_k b_k\right)^2 \leq \left(\sum_{k=1}^n a_k^2\right) \left(\sum_{k=1}^n b_k^2\right)$$



LinuxWindows

Content for Linux...

.NETTypeScriptREST API

.NET content for Linux...